INCH-POUND

MIL-DTL-24439B(SH)
14 July 2026
SUPERSEDING
MIL-V-24439A(SH)
14 March 1984

DETAIL SPECIFICATION

VALVES, OXYGEN, HELIUM, AND HELIUM-OXYGEN MIXTURE – HIGH PRESSURE GAS SERVICES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes valves for high pressure oxygen, helium, and helium-oxygen mixture gas services. The valves are to be suitable for use with an operating system pressure of 4,500 pounds per square inch (lb/in^2).

1.2 Classification. The valves are of the following types and sizes as specified (see 6.2).

1.2.1 Types.

a. Type A - Angle body.
b. Type B - Globe body.
c. Type C - Check, globe body.
d. Type D - Check, angle body.

1.2.2 Sizes.

a. ¼ inch
b. ⅜ inch
c. ½ inch
d. ¾ inch
e. 1 inch
f. 1½ inch
g. 1¾ inch
h. 2 inches
i. 2½ inches

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 4820

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503 - Nitrogen, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-T-1368 - Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded

MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service

MIL-PRF-25567 - Leak Detection Compound, Oxygen Systems

MIL-PRF-27617 - Grease, Aircraft and Instrument, Fuel and Oxidizer Resistant

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

MIL-STD-1474 - Noise Limits

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). These publications are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B124/B124M - Standard Specification for Copper and Copper Alloy Forging Rod, Bar, and Shapes

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B169/B169M - Standard Specification for Aluminum Bronze Sheet, Strip, and Rolled Bar

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

ASTM B187/B187M - Standard Specification for Copper, Bus Bar, Rod, and Shapes and General Purpose Rod, Bar, and Shapes

ASTM B283/B283M - Standard Specification for Copper and Copper-Alloy Die Forgings (Hot-Pressed)

ASTM D1430 - Standard Classification System for Polychlorotrifluoroethylene (PCTFE) Plastics

ASTM D7194 - Standard Specification for Aerospace Parts Machined from Polychlorotrifluoroethylene (PCTFE)

ASTM G63 - Standard Guide for Evaluating Nonmetallic Materials for Oxygen Service

(Copies of these publications are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS4500 - Copper, Sheet, Strip, and Plate Soft Annealed

(Copies of this publication are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Materials. Materials shall be as specified in table I. When not specifically noted, commercial materials suitable for the service intended shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

TABLE I. Materials.

Part	Type A and B	Type C and D	Applicable document
Nipple	Nickel-copper	Nickel-copper	MIL-T-1368
Body	Nickel-copper	Nickel-copper	QQ-N-281
Bonnet	Nickel-copper Bronze	Nickel-copper Bronze	QQ-N-281, QQ-N-286, MIL-B-24480, ASTM B150/B150M
Union nut	Nickel-copper Bronze	Nickel-copper Bronze	QQ-N-281, QQ-N-286 MIL-B-24480, ASTM B150/B150M
Disc	Nickel-copper Bronze	Nickel-copper	QQ-N-281, QQ-N-286
Synthetic insert	Polychlorotrifluoroethylene (PCTFE) polymer	PCTFE polymer, as approved	ASTM D1430, ASTM D7194 1/
Bellows	Nickel-copper Electrodeposited nickel	--- ---	QQ-N-281 ---
Wetted stem assembly	Nickel-copper	---	QQ-N-281, QQ-N-286
External seal	Annealed copper	Annealed copper	SAE AMS4500 ASTMB187/B187M, alloy 110
Handwheel	Aluminum	---	Commercial
Other non-wetted parts	Corrosion-resisting steel (CRES) Bronze	CRES	Commercial

NOTE:

1/ ASTM D1430, group 01, class 1, grade 3 shall be used for items made with resin material. ASTM D7194, type I shall be used for machined parts.

3.2.1 Non-metallic materials. Non-metallic materials in contact with the line fluid shall have a minimum spontaneous ignition temperature of 500 °F as specified by the autoignition requirements of ASTM G63.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction.

3.3.1 Pressure rating. Valves shall be designed for a system operating pressure of 4,500 lb/in^2.

3.3.2 Valve types A and B. Type A and type B valves shall be of the packless, positive shut-off type, with the stem actuated through a bellows. Two to three complete turns of the valve handle shall be required to operate the valve from fully open to fully closed. The handwheel shall turn in the clockwise direction to close the valve. The valve shall employ either a union bonnet or bolted bonnet construction. The valve shall permit full flow in either direction without any tendency for the valve to automatically close. The valve shall operate smoothly throughout its full range with no sudden uncontrolled changes in flow rate for flow in either direction. Valve orifice sizes shall be in accordance with table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

TABLE II. Orifice size.

Valve size (inch)	Minimum orifice diameter size (inch)
¼ to ½	0.20
¾ to 1	0.50
1½ to 2½	1.00

3.3.3 Valve types C and D. Type C and type D valves shall be lift check type and shall be body guided and designed to preclude bypass of the check feature in the reverse flow direction.

3.3.4 Seat leakage. Type A and type B valves shall have a maximum seat leakage of 4×10^{-3} cubic centimeters (cm^3) per second per inch orifice size when pressurized to 4,500 lb/in^2 below the seat and exhausting to atmospheric pressure and 75±5 °F without exceeding the maximum handwheel torques listed in table III, and shall operate properly when installed in any position. Type C and type D valves shall have a maximum seat leakage of 4×10^{-3} cm^3 per second per inch of orifice size when pressurized to 4,500 lb/in^2 in the reverse flow direction and exhausting to atmospheric pressure and 75±5 °F.

TABLE III. Maximum torque for valve seating operation.

Orifice diameter size (inch)	Maximum operating torque (inch/pounds)
0.20	50
0.50	225
1.00	750

3.3.5 Actuating. The design of the valve actuating mechanism shall be arranged such that the valve will not be damaged when the handwheel is torqued to the value permitted in table IV. This shall be accomplished by designing the seat and the actuating mechanism with sufficient strength to accommodate the overload condition.

TABLE IV. Torque values.

Orifice diameter size (inch)	Torque (inch/pounds)
0.20	240
0.50	300
1.00	1,000

3.3.6 Handwheel. The handwheel shall be round with knurling, dimples, or other irregularities on the periphery to provide a firm grip. Spoked wheels may be used on the larger sizes, but the spokes shall not protrude beyond the rim of the wheel. The connection between the handwheel and the valve stem shall provide a positive drive to the stem in either direction. Connections depending solely upon friction or set screws to transmit the torque shall not be acceptable.

3.3.7 Cartridge. The valve shall be of the replaceable cartridge type in which the critical sealing and operating parts may be removed for examination or replacement without removing the valve body from the piping system. No grinding, machining, or welding shall be required for this operation. The replaceable cartridge assembly for each size of valve shall be usable in either type A or type B valve bodies.

3.3.8 Seals. The valve seat, bonnet, and any other seals used shall be arranged to prevent leakage of helium. The primary external seal shall be annealed copper in accordance with table I. Metallic seals shall seat in any direction. Synthetic non-metallic sealing materials shall be as specified in table I and shall be contained in such a manner that they will not extrude or cold flow and allow leakage under system pressure at any temperature between

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

-65 and +160 °F. The seat design shall be such that if the non-metallic seat is destroyed or carried away, there will be a secondary metal-to-metal seat.

3.3.9 Lubrication. No lubrication shall be used within the normal flow path of the system fluid. Threads, thrust bearings, and other moving parts located outside of the primary seals may be lubricated with grease in accordance with MIL-PRF-27617, type III. Lubricants containing petroleum-based materials shall not be acceptable for this service.

3.3.10 Shock. Valves shall withstand the grade A, class I, type B shock test requirement in accordance with MIL-DTL-901 (see 4.3.11).

3.3.11 Vibration. Valves shall withstand the vibration requirements in accordance with MIL-STD-167-1 for type I equipment (see 4.3.12).

3.3.12 Structureborne and airborne noise. Valves shall meet the noise requirements in accordance with MIL-STD-1474 and table V herein (see 4.3.13).

TABLE V. Sound pressure acceptance levels for machinery and equipment in decibels relative to 20 micropascals.

Noise grade	Octave band center frequency (hertz)								
D	31.5	63	125	250	500	1,000	2,000	4,000	8,000
	91	88	85	82	79	76	73	70	67

3.3.13 Pipe nipples or extensions. Pipe nipples or extensions of nickel-copper alloy in accordance with MIL-T-1368 shall be welded to the valve body inlet and outlet. The length and schedule of these extensions shall be as specified in table VI. Welding shall be in accordance with S9074-AR-GIB-010/278 and nondestructive testing shall be in accordance with T9074-AS-GIB-010/271.

TABLE VI. Length and schedule of extension nipples.

Size of valve (inches)	Pipe schedule	Minimum length of extension (center to end) (inches)
¼ to 1 (inclusive)	80	7.0
1½ to 2½ (inclusive)	160	12.0

3.3.14 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.3.15 Endurance. Valves shall be compatible with the maintenance schedule or service life of the installed system (see 4.3.6).

3.4 Identification of product. Each valve shall be permanently and legibly marked with the following:

a. “VALVE, GAS SERVICE, HIGH PRESSURE”.
b. Specification “MIL-DTL-24439”.
c. Manufacturer’s name or trademark.
d. Manufacturer’s part number.
e. National stock number (if known).
f. Nominal pipe size of valve.
g. Component identification (CID) number or allowance parts list (APL) (if known).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with test conditions specified in 4.2.1 and 4.2.2.

4.2.1 High pressure gas. Unless otherwise specified herein, the high pressure gas used in conformance testing and qualification testing shall be water-pumped nitrogen in accordance with A-A-59503, or air equivalent in dryness to A-A-59503 nitrogen, or commercially pure helium, as a minimum.

4.2.2 Test temperature. Unless otherwise specified (see 6.2), all tests except 4.3.7 shall be conducted at a room temperature of 75±5 °F.

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall consist of the examinations of 4.3.3 and 4.3.14 and tests specified in 4.3.4 through 4.3.13 conducted in the order listed (see table VII). Test equipment shall be acceptable to NAVSEA.

TABLE VII. Qualification and conformance inspections.

Characteristic	Qualification	Conformance	Requirement	Verification
Materials, workmanship, design, construction, and marking	X	---	3.2, 3.3, 3.4	4.3.3, 4.3.14
Shell pressure	X	X	3.3.1	4.3.4, 4.4.1
Operating torque	X	---	3.3.2, 3.3.3	4.3.5
Endurance	X	---	3.3.15	4.3.6
Elevated and low temperature	X	---	3.3.8	4.3.7
Full flow	X	---	3.3.2	4.3.8
Seat and bonnet tightness	X	X	3.3.4, 3.3.8	4.3.9, 4.4.2
Pressure retention	X	---	3.3.1, 3.3.4	4.3.10
Shock	X	---	3.3.10	4.3.11
Vibration	X	---	3.3.11	4.3.12
Structureborne and airborne noise	X	---	3.3.12	4.3.13

4.3.1 Sampling. Qualification test samples for each type shall consist of two valves of each size.

4.3.2 Extension of qualification. Qualification approval of the type A or type B valves will qualify all other types provided the same body or a scaled variation of the body is used for type C and type D and that the covers of type C and type D valves are simplifications of the bonnet assemblies used for type A and type B valves. See table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

TABLE VIII. Valve sizes required for qualification and extension of qualification.

Test size (inches)	Extends qualification for these sizes (inches)
2½	2, 1¾, 1½
1	¾
½	¼, ⅜

4.3.3 Examination of product. Each valve shall be examined, without disassembly, to determine conformance with this specification with respect to materials, workmanship, design, construction, and marking.

4.3.4 Shell pressure test. Helium at 6,750 lb/in^2 shall be applied to the valve with the outlet capped and the valve in the open position. Leakage detection compound for oxygen systems in accordance with MIL-PRF-25567, type I shall be applied to body-to-bonnet external leak paths. The duration of the test shall be 5 minutes. There shall be no evidence of distortion or leakage during this time.

4.3.5 Operating torque. Opening and closing torques shall be measured with 4,500 lb/in^2 pressure applied to the valve in each of the following conditions:

a. Pressure applied below the seat with the outlet capped.

b. Pressure applied above the seat with the outlet open. Valve shall be closed bubble tight.

c. Pressure applied below the seat with the outlet open. Valve shall be closed bubble tight.

The torque required to operate the valve shall not exceed the limits specified in table III.

4.3.6 Endurance. The valve shall be installed in a pipeline supplied with 4,500 lb/in^2 pressure and cycle tested. Each cycle shall consist of opening the valve through its complete travel and returning it again to the closed position. The operation shall be smoothly and uniformly performed.

A total of 5,000 complete open-to-close cycles shall be performed at a rate of three to five cycles per minute. The outlet pressure shall be allowed to bleed off through the seat between each cycle. Complete closure of the valve shall be indicated by the cessation of exhaust flow before starting the next cycle. Pressure at the valve seat shall be as follows:

a. System pressure applied under the valve seat for 2,500 cycles.

b. System pressure applied above the valve seat for 2,500 cycles.

There shall be no malfunction and, after each 500 cycles, the valve shall be inspected for bonnet leakage, seat leakage, and increase in operating torque.

4.3.7 Elevated and low temperature tests. The valves shall be tested for operating torques and complete closure (bubble tight) at temperatures of 160±2 °F and -65±2 °F. Pressure of 4,500 lb/in^2 shall be applied to the valve with the outlet open or connected to an exhaust line and the valve closed. The valve shall be conditioned for 1 hour at the respective temperatures. The valve shall be tested by partially opening the valve (moderate flow) and closing the valve for five open-close cycles at each temperature condition. Complete closure (bubble tight) shall be attained at the end of each cycle and the operating torque shall not exceed the allowable torque (see table III) at any time during the cycle.

4.3.8 Full flow. Stop valves shall be tested to ensure that full flow in both directions is obtained through the valve as specified in 3.3.1. There shall be no tendency for the valve to close with the flow in either direction. Check valves shall not fail to close in the reverse flow direction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

4.3.9 Seat and bonnet tightness. The valve shall be pressurized with helium at a minimum of 4,500 lb/in^2 alternately on each side of the seat with the valve closed and the torque not to exceed the values specified in table III, with the other port connected to a bubble-o-meter leak detector. Seat leakage shall be in accordance with 3.3.4. Helium at 4,500 lb/in^2 shall be applied to the valve in the open position with the other port capped. Leakage detection compound for oxygen systems in accordance with MIL-PRF-25567, type I shall be applied to body-to-bonnet external leakage paths. The duration of the test shall be a minimum of 5 minutes. There shall be no evidence of external leakage during this time.

4.3.9.1 Operating torque. The seat tightness test of 4.3.9 shall be accomplished without exceeding the torque values specified in table III. Excessive torque requirements shall be cause for rejection.

4.3.10 Pressure retention. The valve shall be pressurized with helium as specified in 4.2.1 at 4,500 lb/in^2 applied below the valve seat and the valve closed with the torque not to exceed the values specified in table III, then inspected for leakage (bubble tight). A pressure gage, manometer, or other device shall be attached directly to the valve outlet to indicate any leakage past the seat. These conditions shall be held continuously for a period of 5 days. During this period, the maximum leakage rate shall be in accordance with 3.3.4. The ambient temperature at the start of the test shall be used as the reference temperature. The pressurized helium shall be stabilized at the ambient condition, and the reading shall be corrected to the starting condition at the end of the test.

4.3.11 Shock. High-impact shock tests shall be conducted in accordance with the requirements of MIL-DTL-901, grade A, class 1, type B as follows:

a. The valve shall perform identically before and after the test.

b. Valves shall be installed with approximately 2 inches of pipe between the valve and the supporting bracket. Breakage of the pipe shall constitute failure of the valve.

c. The valve shall be in the closed position when shock tested.

d. The valve shall be hydrostatically pressurized on the upstream side at the design pressure of 4,500 lb/in^2.

e. Following tests and examination, the valve shall be retained for disposition.

f. No exceptions to the above requirements and MIL-DTL-901 shall be permitted.

g. Testing of items from each regular production run shall not be required.

4.3.12 Vibration. Valves shall be subjected to and pass the type I environmental vibration test in accordance with MIL-STD-167-1.

4.3.13 Structureborne and airborne noise. Valves shall be subjected to and pass the grade D and type III noise test in accordance with MIL-STD-1474. The system flow conditions shall be 29 standard cubic feet per minute minimum per inch of orifice size.

4.3.14 Visual examination. Upon completion of the tests specified in 4.3.4 through 4.3.13, the valves shall be completely disassembled and all parts inspected to determine wear, distortion, or other damage. The valve parts shall not be excessively worn or damaged.

4.4 Conformance inspection. Each valve shall be tested as specified in 4.4.1 and 4.4.2 (see table VII).

4.4.1 Shell pressure test. Each valve shall be proof tested with nitrogen air or helium (see 4.2.1) at a minimum pressure of 6,750 lb/in^2. There shall be no visual evidence of leakage or distortion (see 4.3.4).

4.4.2 Seat and bonnet tightness. Seat and bonnet tightness shall be tested as specified in 4.3.9 and 4.3.9.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. This specification covers valves to be used on high pressure gas services including such gases as oxygen, helium, and helium-oxygen mixtures. The valve is designed for services in which external leakage must be maintained at an absolute minimum seat tightness sustained for gases of small molecular structure and services requiring clean piping systems. Valves procured to this specification are form, fit, and function replacements for prior revisions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Test temperature if other than specified (see 4.2.2).

d. Packaging requirements (see 5.1).

e. Furnishing of Provisioning Technical Documentation (PTD), spare parts, and repair parts (see 6.5).

f. Supplemental requirements for special sizes, capacities, and applications.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24439 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Provisioning. PTD, spare parts, and repair parts should be furnished as specified (see 6.2). When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Subject term (key word) listing.

Cartridge style valves
Gas valves
Globe valves
Needle valves
Throttling valves

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CC

Preparing activity:
Navy – SH
(Project 4820-2026-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:54Z
Check the source to verify that this is the current version before use.